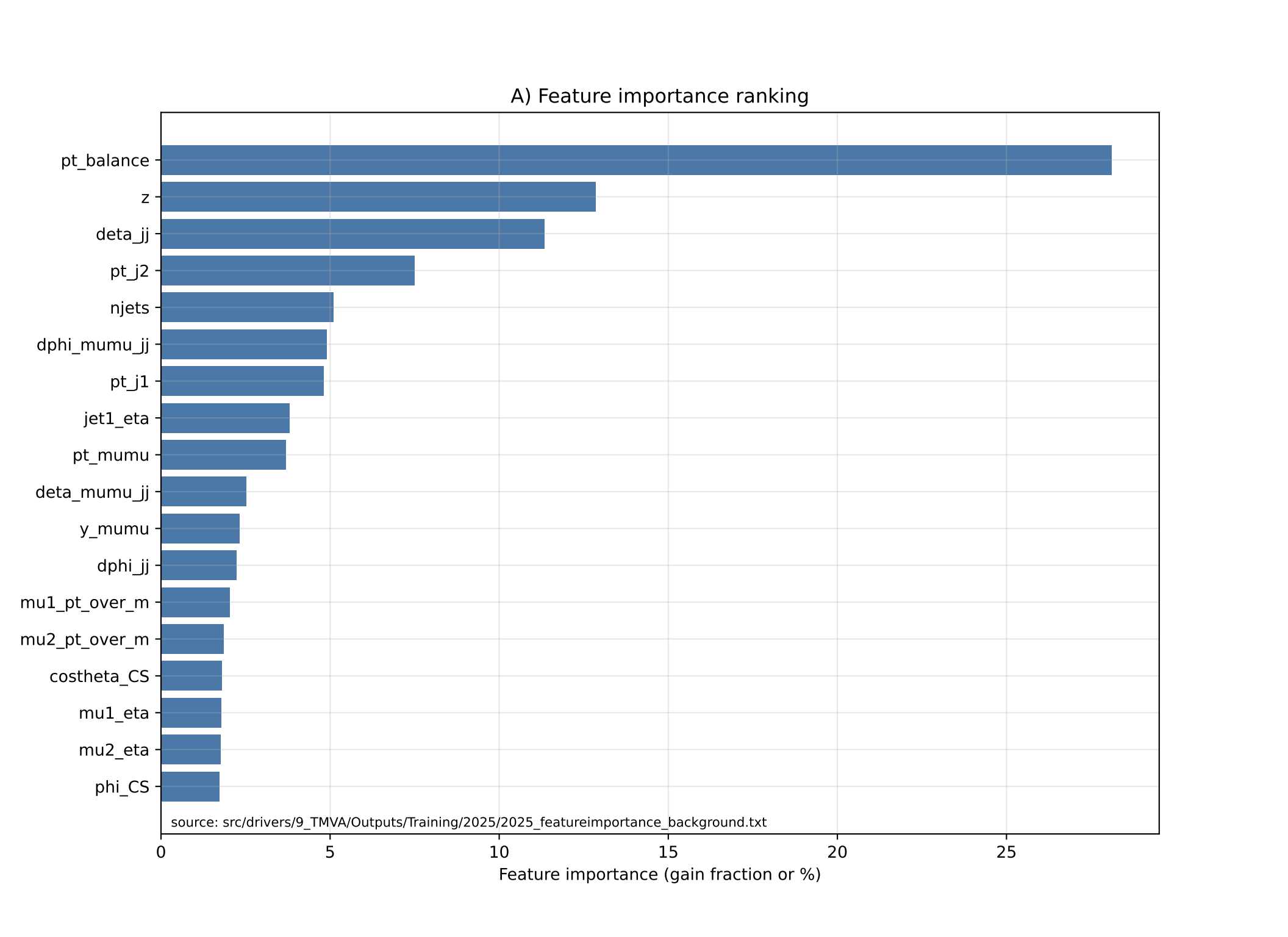
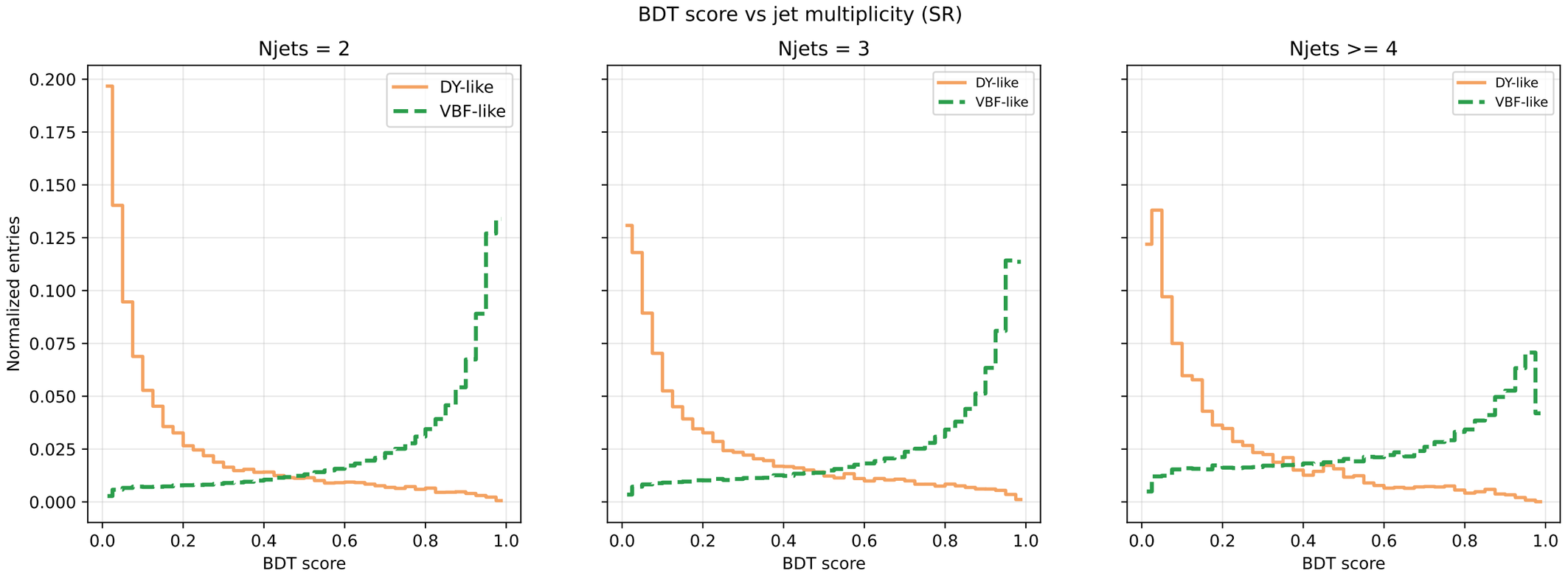
VBF BDT Validation - 2025

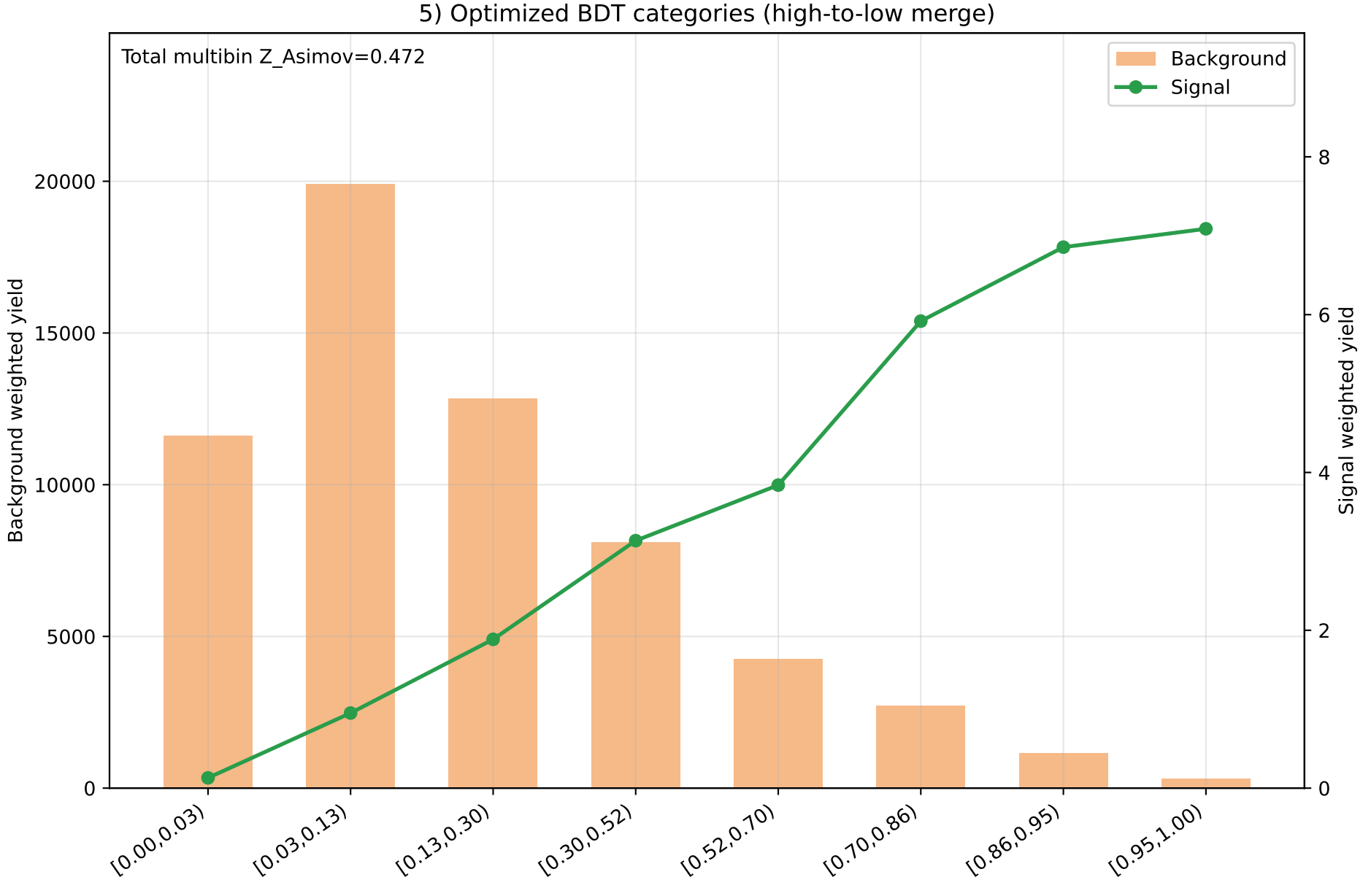
```
Standalone per-era diagnostic script: bdt_validation.py
Logic references: ScanYields / PlotScored / MassSculpt / KS_perfold / DataMC
Resolved branches: {'score': 'bdt_score', 'weight': 'weight', 'mass': 'm_mumu', 'njets': 'njets', 'mjj': None, 'de
```

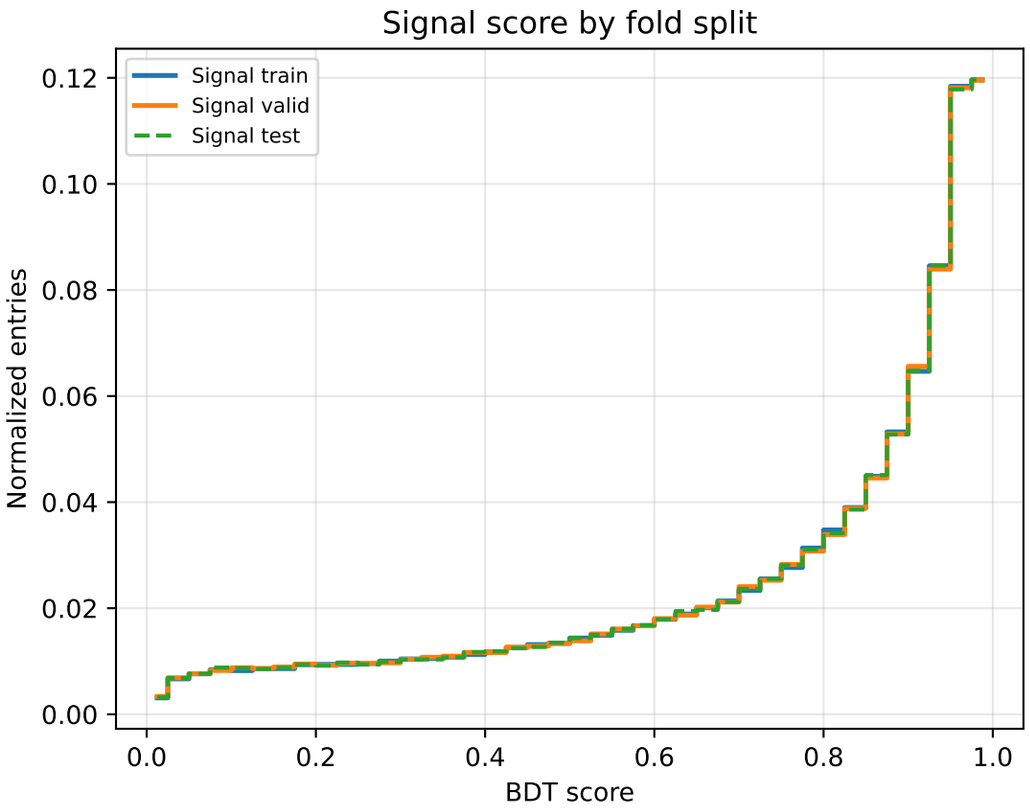
Requested VBF feature cross-check

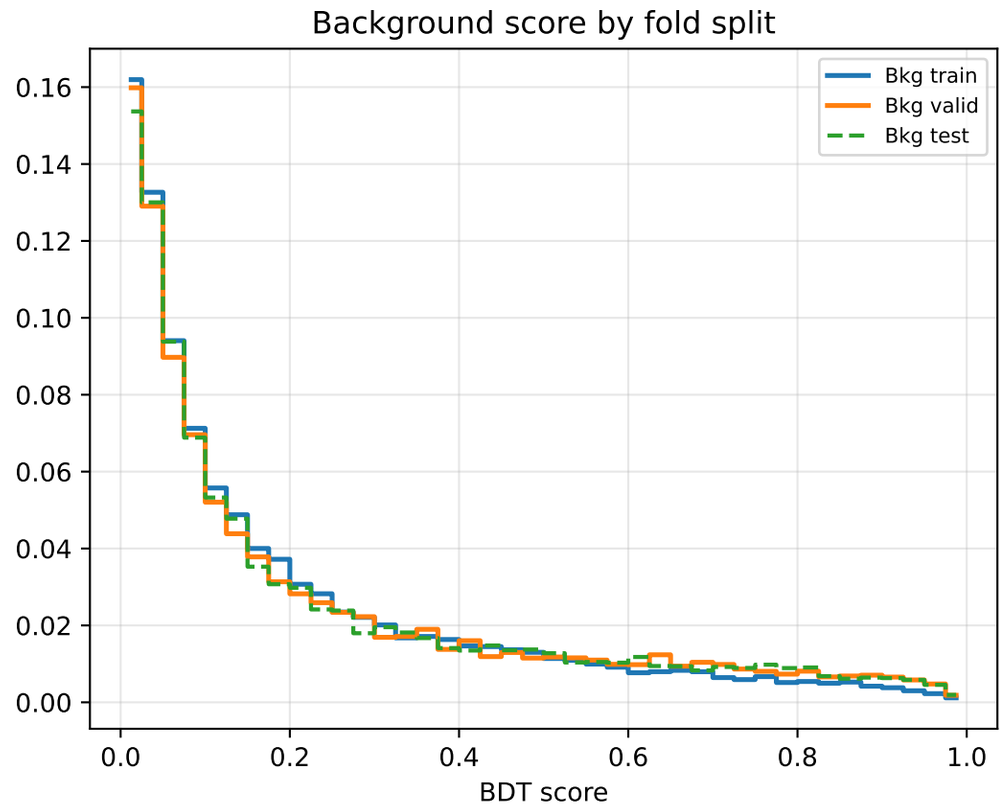
requested	status	resolved_as	notes
jet1_qgl	missing	-	not found in tree branches
jet1_eta	exact	jet1_eta	
jet2_eta	proxy	y_j2	rapidity, not eta
jet1_pt	alias	pt_j1	
jet2_pt	alias	pt_j2	
mjj	missing	-	not found in tree branches
dEta_jj	alias	deta_jj	
dPhi_jj	alias	dphi_jj	
mmj_min_dEta	proxy	deta_mumu_jj	dimuon-dijet deta, not min(mu,jet)
mmj_min_dPhi	proxy	dphi_mumu_jj	dimuon-dijet dphi, not min(mu,jet)
pt_centrality	proxy	z	zeppenfeld variable, not pt centrality
dimuon_pt	alias	pt_mumu	
dimuon_rapidity	alias	y_mumu	
dimuon_cos_theta_cs	alias	costheta_CS	
dimuon_phi_cs	alias	phi_CS	
dimuon_mass_resolution	missing	-	not found in tree branches
mu1_pt_over_mass	alias	mu1_pt_over_m	
mu2_pt_over_mass	alias	mu2_pt_over_m	
mu1_eta	exact	mu1_eta	
mu2_eta	exact	mu2_eta	
N_jets	alias	njets	
year	derived	year=2025	from --era


A) Feature importance ranking
pt_balance
z
deta_jj
pt_j2
njets
dphi_mumu_jj
pt_j1
jet1_eta
pt_mumu
deta_mumu_jj
y_mumu
dphi_jj
mu1_pt_over_m
mu2_pt_over_m
costheta_CS
mu1_eta
mu2_eta
phi_CS
source: src/drivers/9_TMVA/Outputs/Training/2025/2025_featureimportance_background.txt
0
5
10
15
20
25
Feature importance (gain fraction or %)


BDT score vs jet multiplicity (SR)
Njets = 2
Njets = 3
Njets >= 4
DY-like
VBF-like
Normalized entries
BDT score
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.0
0.2
0.4
0.6
0.8
1.0


5) Optimized BDT categories (high-to-low merge)
Total multibin Z_Asimov=0.472
Background
Signal
Background weighted yield
Signal weighted yield
0
5000
10000
15000
20000
0
2
4
6
8
[0.00,0.03)
[0.03,0.13)
[0.13,0.30)
[0.30,0.52)
[0.52,0.70)
[0.70,0.86)
[0.86,0.95)
[0.95,1.00)



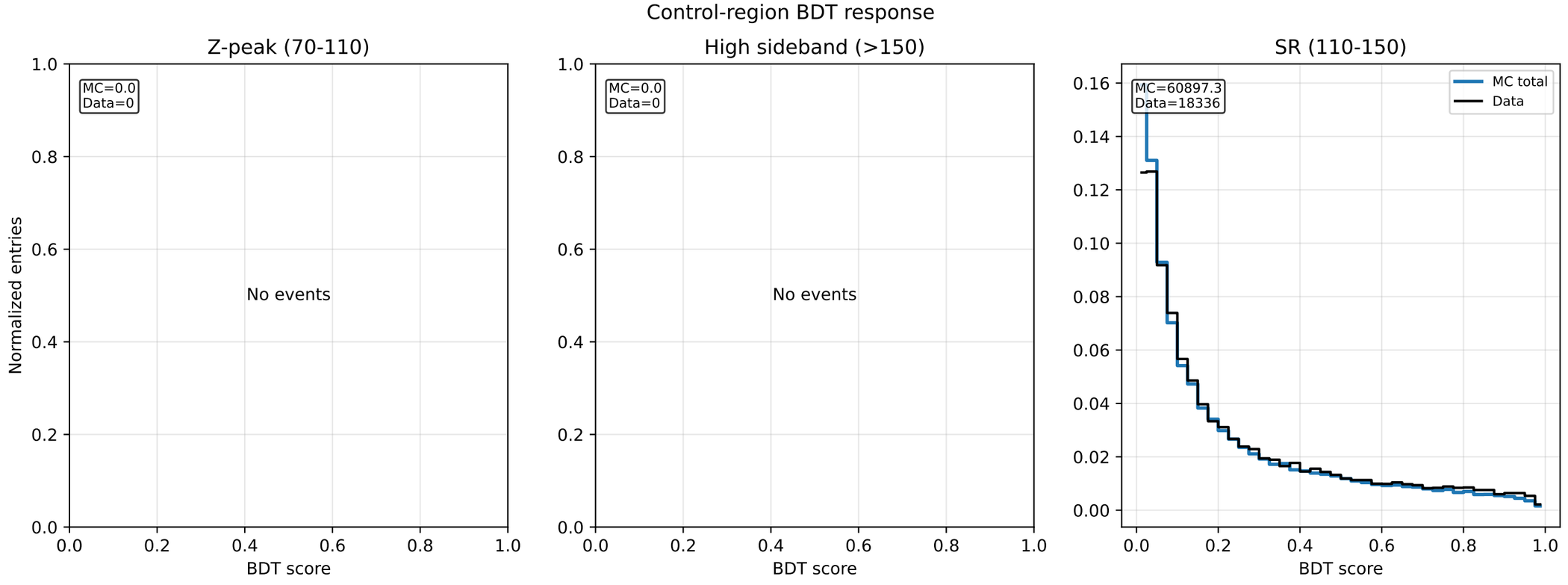


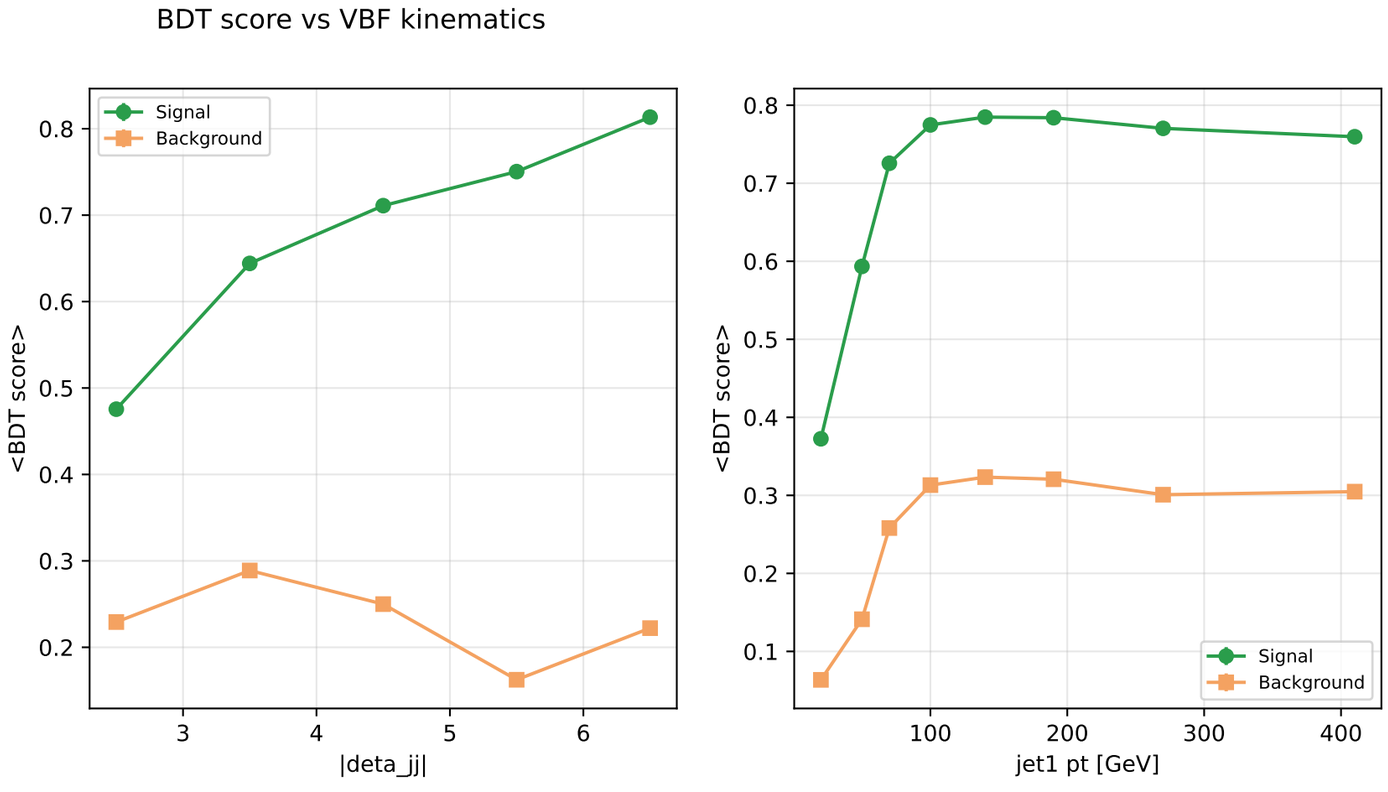
Fold-based KS checks (train vs validation)

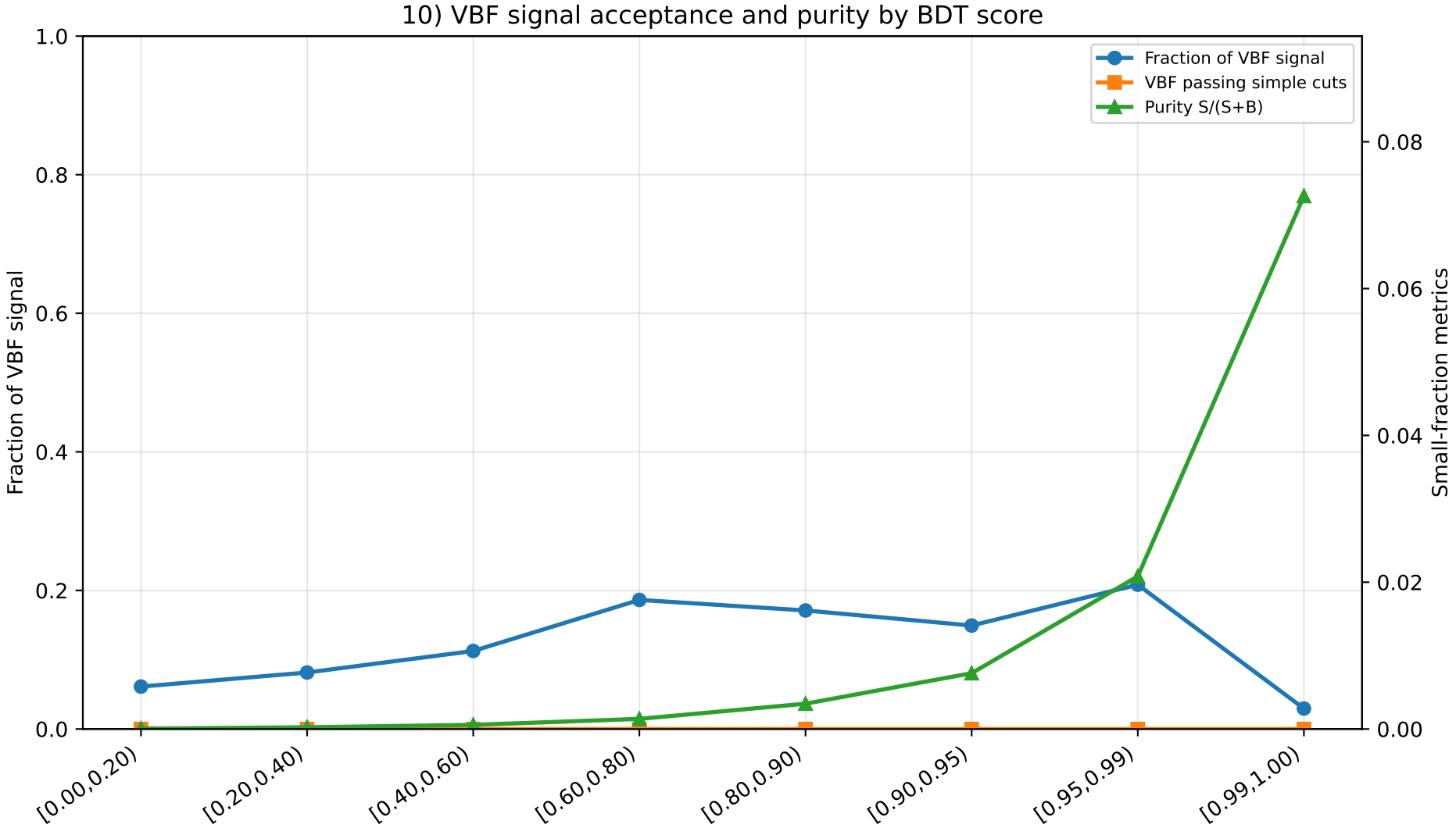
KS thresholds: Run2-loose: p>=0.01, D<=0.15; Run2-nominal: p>=0.05, D<=0.1; Run2-tight: p>=0.1, D<=0.07

Sample	N train	N valid	N test	Mean(train)	Mean(valid)	Mean(test)	KS D	KS p	Run2-loose	Run2-nominal	Run2-tight
Signal	800568	400284	400284	0.7381	0.7369	0.7371	0.0012	0.8527	**PASS**	**PASS**	**PASS**
Background	219518	109754	109752	0.2087	0.2327	0.2333	0.0067	0.002602	**FAIL**	**FAIL**	**FAIL**

Corr(score, njets): signal=-0.142, background=0.022


Control-region BDT response
Z-peak (70-110)
High sideband (>150)
SR (110-150)
MC=0.0
Data=0
No events
MC=0.0
Data=0
No events
MC=60897.3
Data=18336
MC total
Data
Normalized entries
BDT score
BDT score
BDT score


BDT score vs VBF kinematics
Signal
Background
<BDT score>
|deta_jj|
jet1 pt [GeV]


10) VBF signal acceptance and purity by BDT score
Fraction of VBF signal
VBF passing simple cuts
Purity S/(S+B)
Fraction of VBF signal
Small-fraction metrics
1.0
0.8
0.6
0.4
0.2
0.0
0.08
0.06
0.04
0.02
0.00
[0.00,0.20)
[0.20,0.40)
[0.40,0.60)
[0.60,0.80)
[0.80,0.90)
[0.90,0.95)
[0.95,0.99)
[0.99,1.00)

Validation Summary

Era: 2025
Inputs: data_files=1 mc_groups=7
Signal groups: SIGNAL
VBF groups: SIGNAL
SR weighted yields: S=29.804, B=60867.453
KS profiles: Run2-loose: p>=0.01, D<=0.15; Run2-nominal: p>=0.05, D<=0.1; Run2-tight: p>=0.1, D<=0.07
Feature importance source: src/drivers/9_TMVA/Outputs/Training/2025/2025_featureimportance_background.txt
Optimized categories: n=8, Z_Asimov=0.4720
KS(signal train-vs-valid): D=0.0012, p=0.8527
KS(bkg train-vs-valid): D=0.0067, p=0.002602
KS profile checks (Signal): Run2-loose=PASS, Run2-nominal=PASS, Run2-tight=PASS
KS profile checks (Background): Run2-loose=FAIL, Run2-nominal=FAIL, Run2-tight=FAIL
Corr(score, njets) signal=-0.142, background=0.022
Highest BDT-bin purity=0.073